

SAFETY DATA SHEET

Creation Date 30-December-2014

Revision Date 24-December-2021

Revision Number 5

1. Identification

Product Name FL-70 Detergent, Biodegradable

Cat No. : SF105-1; SF105-4

Synonyms Concentrated cleaning compound

Recommended Use Laboratory chemicals.

Uses advised against Food, drug, pesticide or biocidal product use.

Details of the supplier of the safety data sheet

Company

Importer/Distributor
Fisher Scientific
112 Colonnade Road,
Ottawa, ON K2E 7L6,
Canada
Tel: 1-800-234-7437

Manufacturer

Fisher Scientific Company
One Reagent Lane
Fair Lawn, NJ 07410
Tel: (201) 796-7100

Emergency Telephone Number CHEMTREC®, Inside the USA: 800-424-9300
CHEMTREC®, Outside the USA: 001-703-527-3887

2. Hazard(s) identification

Classification

WHMIS 2015 Classification Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

Serious Eye Damage/Eye Irritation Category 2

Label Elements

Signal Word

Warning

Hazard Statements

Causes serious eye irritation

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling

Wear eye/face protection

Response

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

If eye irritation persists: Get medical advice/attention

Disposal

Dispose of contents/container to an approved waste disposal plant

3. Composition/Information on Ingredients

Component	CAS-No	Weight %
Water	7732-18-5	88.8
9-Octadecenoic acid (9Z)-, compound with 2,2,2-nitrilotris[ethanol] (1:1)	2717-15-9	3.8
Disodium carbonate	497-19-8	2.7
Alcohols, C12-14-secondary, ethoxylated	84133-50-6	1.8
Tetrasodium EDTA	64-02-8	1.4
Polyethylene glycol	25322-68-3	0.9
Sodium oleate	143-19-1	0.5
Sodium bicarbonate	144-55-8	0.1

4. First-aid measures

General Advice	If symptoms persist, call a physician.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.
Inhalation	Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.
Ingestion	Clean mouth with water and drink afterwards plenty of water.
Most important symptoms/effects	None reasonably foreseeable.
Notes to Physician	Treat symptomatically

5. Fire-fighting measures

Suitable Extinguishing Media	Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.
Unsuitable Extinguishing Media	No information available

Flash Point	No information available
Method -	No information available
Autoignition Temperature	No information available
Explosion Limits	
Upper	No data available
Lower	No data available
Sensitivity to Mechanical Impact	No information available
Sensitivity to Static Discharge	No information available

Specific Hazards Arising from the Chemical

Non-combustible. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

Hazardous Combustion Products

Thermal decomposition can lead to release of irritating gases and vapors.

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

NFPA

Health
2

Flammability
0

Instability
0

Physical hazards
N/A

6. Accidental release measures

Personal Precautions

Use personal protective equipment as required. Ensure adequate ventilation.

Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

Methods for Containment and Clean Up Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

7. Handling and storage

Handling

Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

Storage.

Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. No information available.

8. Exposure controls / personal protection

Exposure Guidelines

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment**Eye Protection**

Goggles

Hand Protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Glove material	Breakthrough time	Glove thickness	Glove comments
Natural rubber	See manufacturers	-	Splash protection only
Nitrile rubber	recommendations		
Neoprene			
PVC			

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Recommended Filter type: Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls

No information available.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

9. Physical and chemical properties

Physical State	Liquid
Appearance	Yellow
Odor	Odorless
Odor Threshold	No information available
pH	12.0 Alkaline
Melting Point/Range	0 °C / 32 °F
Boiling Point/Range	> 100 °C / 212 °F
Flash Point	No information available
Evaporation Rate	> 1 (Ether = 1.0)
Flammability (solid,gas)	Not applicable
Flammability or explosive limits	
Upper	No data available
Lower	No data available
Vapor Pressure	No information available
Vapor Density	> 1.0
Specific Gravity	1.04
Solubility	Soluble in water
Partition coefficient; n-octanol/water	No data available
Autoignition Temperature	No information available
Decomposition Temperature	No information available
Viscosity	No information available

10. Stability and reactivity

Reactive Hazard	None known, based on information available
Stability	Stable under normal conditions.
Conditions to Avoid	Excess heat.
Incompatible Materials	No information available

Hazardous Decomposition Products Thermal decomposition can lead to release of irritating gases and vapors

Hazardous Polymerization Hazardous polymerization does not occur.

Hazardous Reactions None under normal processing.

11. Toxicological information

Acute Toxicity

Product Information

Oral LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

Dermal LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

Vapor LC50

Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

Component Information

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	-	-	-
Disodium carbonate	2800 mg/kg (Rat)	> 2000 mg/kg (rabbit)	2.3 mg/l 2h (Rat)
Alcohols, C12-14-secondary, ethoxylated	LD50 = 2100 mg/kg (Rat)	Not listed	Not listed
Tetrasodium EDTA	LD50 = 1780 - 2000 mg/kg (Rat)	Not listed	Not listed
Polyethylene glycol	LD50 = 22 g/kg (Rat)	LD50 > 20 g/kg (Rabbit)	Not listed
Sodium bicarbonate	LD50 = 4220 mg/kg (Rat)	Not listed	Not listed

Toxicologically Synergistic Products No information available

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Irritation Irritating to eyes

Sensitization No information available

Carcinogenicity The table below indicates whether each agency has listed any ingredient as a carcinogen.

Component	CAS-No	IARC	NTP	ACGIH	OSHA	Mexico
Water	7732-18-5	Not listed	Not listed	Not listed	Not listed	Not listed
9-Octadecenoic acid (9Z)-, compound with 2,2,2-nitritotris[ethanol] (1:1)	2717-15-9	Not listed	Not listed	Not listed	Not listed	Not listed
Disodium carbonate	497-19-8	Not listed	Not listed	Not listed	Not listed	Not listed
Alcohols, C12-14-secondary, ethoxylated	84133-50-6	Not listed	Not listed	Not listed	Not listed	Not listed
Tetrasodium EDTA	64-02-8	Not listed	Not listed	Not listed	Not listed	Not listed
Polyethylene glycol	25322-68-3	Not listed	Not listed	Not listed	Not listed	Not listed
Sodium oleate	143-19-1	Not listed	Not listed	Not listed	Not listed	Not listed
Sodium bicarbonate	144-55-8	Not listed	Not listed	Not listed	Not listed	Not listed

Mutagenic Effects No information available

Reproductive Effects No information available.

Developmental Effects No information available.

Teratogenicity No information available.

STOT - single exposure None known

STOT - repeated exposure None known

Aspiration hazard No information available

Symptoms / effects, both acute and delayed No information available

Endocrine Disruptor Information No information available

Other Adverse Effects The toxicological properties have not been fully investigated.

12. Ecological information

Ecotoxicity

This product contains the following substance(s) which are hazardous for the environment. .

Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea
Disodium carbonate	Not listed	Lepomis macrochirus: LC50: 300 mg/L/96h Gambusia affinis: LC50: 740 mg/L/96h	-	EC50: = 265 mg/L, 48h (Daphnia magna)
Alcohols, C12-14-secondary, ethoxylated	Not listed	LC50: = 3.2 mg/L, 96h (Pimephales promelas)	Not listed	EC50: = 3.2 mg/L, 48h (water flea)
Tetrasodium EDTA	EC50: = 1.01 mg/L, 72h (Desmodesmus subspicatus)	LC50: = 121 - 1592 mg/L, 96h static (Lepomis macrochirus)	Not listed	EC50: = 140mg/l, 48h (Daphnia magna)
Polyethylene glycol	EC50 > 100 mg/L 96h, (Scenedesmus subspicatus) OECD Guideline 201	LC50 > 100 mg/L 96h, (Poecilia reticulata) OECD Guideline 203	Not listed	EC50 > 100 mg/L 48h, (Daphnia magna) OECD Guideline 202
Sodium bicarbonate	EC50: 650 mg/L/120h	LC50: 8250 - 9000 mg/L, 96h static (Lepomis macrochirus)	-	EC50: 2350 mg/L/48h

Persistence and Degradability Soluble in water Persistence is unlikely based on information available.

Bioaccumulation/ Accumulation No information available.

Mobility Will likely be mobile in the environment due to its water solubility.

13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

14. Transport information

DOT Not regulated
TDG Not regulated
IATA Not regulated
IMDG/IMO Not regulated

15. Regulatory information

International Inventories

Component	CAS-No	DSL	NDSL	TSCA	TSCA Inventory notification - Active-Inactive	EINECS	ELINCS	NLP
Water	7732-18-5	X	-	X	ACTIVE	231-791-2	-	-
9-Octadecenoic acid (9Z)-, compound with 2,2,2-nitritotris[ethanol] (1:1)	2717-15-9	X	-	X	ACTIVE	220-311-7	-	-
Disodium carbonate	497-19-8	X	-	X	ACTIVE	207-838-8	-	-
Alcohols, C12-14-secondary,	84133-50-6	X	-	X	ACTIVE	-	-	-

ethoxylated								
Tetrasodium EDTA	64-02-8	X	-	X	ACTIVE	200-573-9	-	-
Polyethylene glycol	25322-68-3	X	-	X	ACTIVE	-	-	500-038-2
Sodium oleate	143-19-1	X	-	X	ACTIVE	205-591-0	-	-
Sodium bicarbonate	144-55-8	X	-	X	ACTIVE	205-633-8	-	-

Component	CAS-No	IECSC	KECL	ENCS	ISHL	TCSI	AICS	NZIoC	PICCS
Water	7732-18-5	X	KE-35400	X	-	X	X	X	X
9-Octadecenoic acid (9Z)-, compound with 2,2,2-nitritoltris[ethanol] (1:1)	2717-15-9	X	-	X	X	X	X	X	X
Disodium carbonate	497-19-8	X	KE-31380	X	X	X	X	X	X
Alcohols, C12-14-secondary, ethoxylated	84133-50-6	X	KE-00456	X	X	X	X	X	X
Tetrasodium EDTA	64-02-8	X	KE-13654	X	X	X	X	X	X
Polyethylene glycol	25322-68-3	X	KE-20228	X	X	X	X	X	X
Sodium oleate	143-19-1	X	KE-26519	X	X	X	X	X	X
Sodium bicarbonate	144-55-8	X	KE-31360	X	X	X	X	X	X

Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

Component	Canada - National Pollutant Release Inventory (NPRI)	Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances	Canada's Chemicals Management Plan (CEPA)
Polyethylene glycol	Part 4 Substance		

Legend

NPRI - National Pollutant Release Inventory

Other International Regulations**Authorisation/Restrictions according to EU REACH**

Component	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Disodium carbonate	-	Use restricted. See item 75. (see link for restriction details)	-
Tetrasodium EDTA	-	Use restricted. See item 75. (see link for restriction details)	-

<https://echa.europa.eu/substances-restricted-under-reach>

Safety, health and environmental regulations/legislation specific for the substance or mixture

Component	CAS-No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential	Restriction of Hazardous Substances (RoHS)
Water	7732-18-5	Listed	Not applicable	Not applicable	Not applicable
9-Octadecenoic acid (9Z)-,	2717-15-9	Not applicable	Not applicable	Not applicable	Not applicable

compound with 2,2,2-nitritotris[ethanol] (1:1)					
Disodium carbonate	497-19-8	Listed	Not applicable	Not applicable	Not applicable
Alcohols, C12-14-secondary, ethoxylated	84133-50-6	Listed	Not applicable	Not applicable	Not applicable
Tetrasodium EDTA	64-02-8	Listed	Not applicable	Not applicable	Not applicable
Polyethylene glycol	25322-68-3	Listed	Not applicable	Not applicable	Not applicable
Sodium oleate	143-19-1	Not applicable	Not applicable	Not applicable	Not applicable
Sodium bicarbonate	144-55-8	Listed	Not applicable	Not applicable	Not applicable

Component	CAS-No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
Water	7732-18-5	Not applicable	Not applicable	Not applicable	Not applicable
9-Octadecenoic acid (9Z)-, compound with 2,2,2-nitritotris[ethanol] (1:1)	2717-15-9	Not applicable	Not applicable	Not applicable	Not applicable
Disodium carbonate	497-19-8	Not applicable	Not applicable	Not applicable	Not applicable
Alcohols, C12-14-secondary, ethoxylated	84133-50-6	Not applicable	Not applicable	Not applicable	Not applicable
Tetrasodium EDTA	64-02-8	Not applicable	Not applicable	Not applicable	Not applicable
Polyethylene glycol	25322-68-3	Not applicable	Not applicable	Not applicable	Not applicable
Sodium oleate	143-19-1	Not applicable	Not applicable	Not applicable	Not applicable
Sodium bicarbonate	144-55-8	Not applicable	Not applicable	Not applicable	Not applicable

16. Other information

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Revision Summary

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of SDS